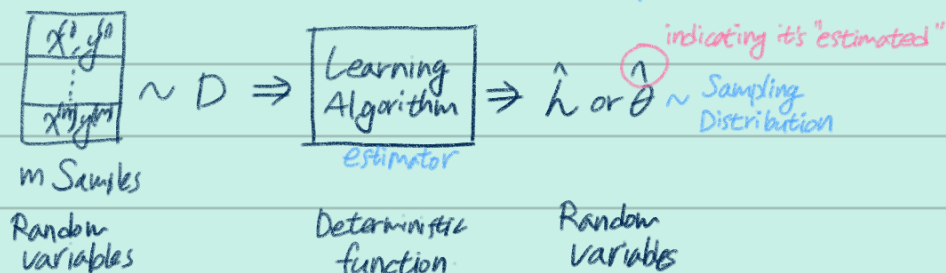


# Lecture 9. Approx/Estimation Error & ERM (Learning Theory)

- Setup / Assumptions
- Bias Variance
- Approx. Estim
- Empirical Risk Minimizer
- Uniform Convergence
- VC dimension

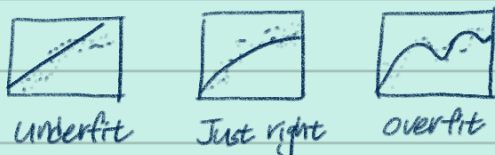
## Assumptions

1. Data distribution  $D$  from which  $(x, y) \sim D$   <sup>$(x, y)$  pairs are sampled</sup> Train Test No probability distribution associated
2. Independent Samples  $\theta^*$  or  $h^*$  "True" parameter Not Random

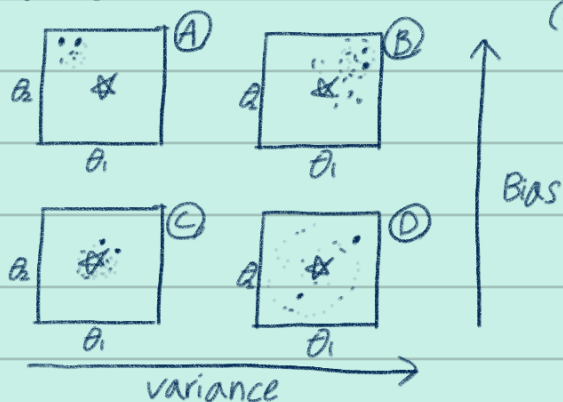


## Bias/Variance

### Data View



### Parameter View

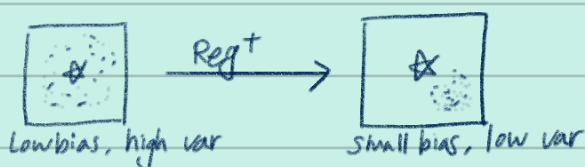


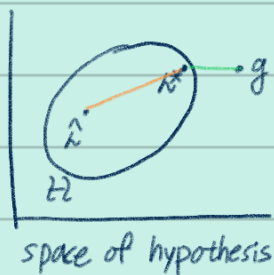
(A ~ D) learning algorithm,  $\star$  is true/unknown parameter

- $m \rightarrow \infty$   
 $\text{Var}[\hat{\theta}] \rightarrow 0$   $\hat{\theta}$ 이 거의 같아짐
- "Statistical Efficiency" rate  
 $\text{Var}[\hat{\theta}] \rightarrow 0$  as  $m \rightarrow \infty$   $O(1/m^2)$   $O(e^{-m})$
- $\hat{\theta} \rightarrow \theta^*$   $m \rightarrow \infty$ : consistency
- $E[\hat{\theta}] = \theta^*$  for all  $m$

## Fighting Variance

1.  $m \rightarrow \infty$
2. Regularization





$g$ : Best possible hypothesis ("True" hypothesis)

$h^*$ : Best in class  $\mathcal{H}$

$\hat{h}$ : Learnt from finite data

$\mathcal{E}(h)$ : Risk / Generalization Error

$$= E(x, y) \sim P[1\{h(x) \neq y\}]$$

sample examples from the data-generating process, fraction of all examples you make a mistake

$\hat{\mathcal{E}}_S(h)$ : Empirical Risk

$$= \frac{1}{m} \sum_{i=1}^m 1\{h(x^{(i)}) \neq y^{(i)}\}$$

$\mathcal{E}(g)$ : Bayes Error / Irreducible error

$\mathcal{E}(h^*) - \mathcal{E}(g)$  = Approximation Error } class

$\mathcal{E}(\hat{h}) - \mathcal{E}(h^*)$  = Estimation Error

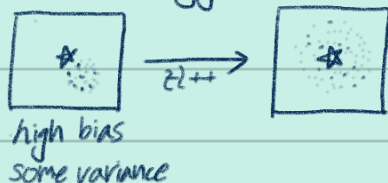
$\mathcal{E}(\hat{h})$  = Estimation Error + Approximation Error + Irreducible Error

Est Var + Est Bias + Approx Error + Irreducible Error

$\mathcal{E}(\hat{h})$  = Var + Bias + Irreducible

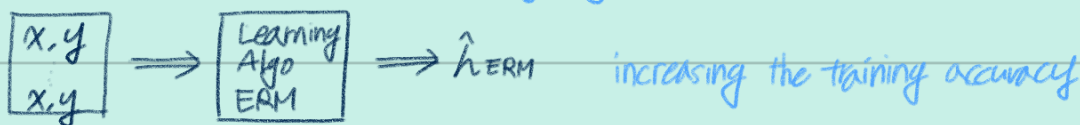
Fighting High Bias

1. make  $\mathcal{H}$  bigger



bias < variance trade off with class size

Empirical Risk Minimizer a learning algorithm



$$\hat{h}_{ERM} = \operatorname{argmin}_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m 1\{h(x^{(i)}) \neq y^{(i)}\}$$

trying to find a minimizer in a class of hypothesis that minimizes the average training error

Uniform Convergence

1.  $\hat{\mathcal{E}}(h)$  vs.  $\mathcal{E}(h)$  empirical risk vs. generalization risk

2.  $\mathcal{E}(\hat{h})$  vs.  $\mathcal{E}(h^*)$

Tools: (1) Union Bound  $A_1, A_2, \dots, A_K$  (need not be indep)

$$P(A_1 \cup A_2 \cup \dots \cup A_K) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_K)$$

## (2) Hoeffding's Inequality

estimator 부분은 개별로 모든 구간을 탐색

Let  $z_1, z_2, \dots, z_m \sim \text{Bern}(\phi)$   $\hat{\phi} = \frac{1}{m} \sum_{i=1}^m z_i$

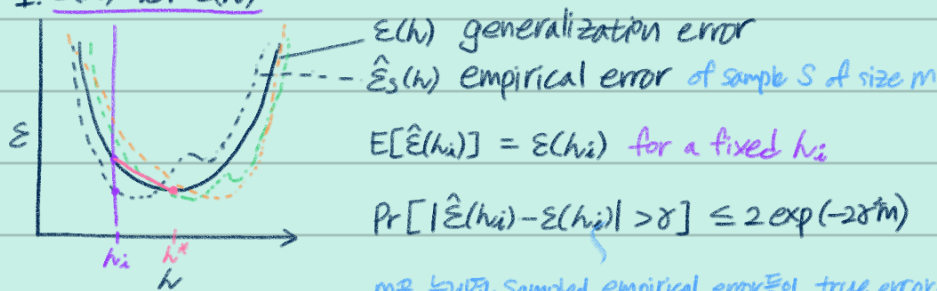
Let  $\delta > 0$  [margin]  $\delta$ 은 우리가 정한 값! e.g. 0.001

$\Pr[|\hat{\phi} - \phi| > \delta] \leq 2 \exp(-2\delta^2 m)$  func of m

Deviation of error margin

Estimate deviating more than a certain margin only reduces as you increase m.

### 1. $\hat{\epsilon}(h)$ vs. $\epsilon(h)$



⇒ Risk curve converges uniformly to the generalization risk curve "uniform convergence"

Finite Hypothesis class  $\mathcal{H}$

$|\mathcal{H}| = K$

$\Pr[\exists h \in \mathcal{H} | \hat{\epsilon}_S(h) - \epsilon(h) > \delta] \leq K \cdot 2 \exp(-2\delta^2 m)$

⇒  $\Pr[\forall h \in \mathcal{H} | \hat{\epsilon}_S(h) - \epsilon(h) < \delta] \geq 1 - \underbrace{2K \cdot \exp(-2\delta^2 m)}_{\delta}$

Let  $\delta = 2K \cdot \exp(-2\delta^2 m)$  probability of error

$\delta$  = margin of the error

$m$  = sample size

Fix  $\delta, \delta > 0$ ,  $m \geq \frac{1}{2\delta^2} \log \frac{2K}{\delta}$  sample complexity dessert

### 2. $\epsilon(\hat{h})$ vs. $\epsilon(h^*)$ Best in class

$\epsilon(\hat{h}) \leq \hat{\epsilon}(\hat{h}) + \delta$

$\leq \hat{\epsilon}(h^*) + \delta$

$\leq \epsilon(h^*) + 2\delta$

⇒ with probability  $1 - \delta$ , training size  $m$

$\epsilon(\hat{h}) \leq \epsilon(h^*) + 2\sqrt{\frac{1}{2m} \log \frac{2K}{\delta}}$

VC Dimension

$VC(\mathcal{H}) = \sim$

$\epsilon(\hat{h}) \leq \epsilon(h^*) + O\left(\sqrt{\frac{VC(\mathcal{H})}{m} \log\left(\frac{m}{VC(\mathcal{H})}\right) + \frac{1}{m} \log \frac{1}{\delta}}\right)$